



Chapter 1: Gravitation

Science & Technology Part 1 / Maharashtra SSC Board

Complete Board Exam Notes – Easy Language



What's Inside These Notes?

1. What is Gravitation?
2. Circular Motion & Centripetal Force
3. Kepler's Three Laws of Planetary Motion
4. Newton's Universal Law of Gravitation
5. Derivation: Relation between g and G
6. Acceleration due to Gravity (g)
7. Variation of ' g ' (Height, Depth, Shape)
8. Mass vs Weight (with Table)
9. Free Fall (with Solved Numericals)
10. Gravitational Potential Energy
11. Escape Velocity
12. 🌟 Board Exam Cheat Sheet & Memory Tricks

💡 **Study Tip:** Read these notes like a story. Every concept is explained so simply that even a 6th class student can understand it. All formulas are highlighted in **orange boxes**. All important definitions are in **yellow boxes**.

🌟 Total Pages: 14 🌟

1. What is Gravitation? 🤔

Definition: The force of attraction between any two objects in the universe is called **gravitational force**. This phenomenon is called **gravitation**.

In Simple Words: Have you ever wondered why:

- 🍎 An apple falls DOWN when you drop it?
- 🌍 The Moon goes around the Earth instead of flying away?
- ☀️ The Earth goes around the Sun?

The answer is: Because of an **invisible pulling force** called **Gravitation**!

Every single thing in the universe – from a tiny ant to a giant star – pulls every other thing towards itself. You are pulling your chair, your chair is pulling you! But because you both are very light compared to the Earth, you don't feel it.

⚠️ **Remember:** Gravity and Gravitation are slightly different!

- **Gravitation** = Force between ANY two objects in the universe.
- **Gravity** = The special case when one of the objects is the **Earth**.

2. Circular Motion 🎡

Definition: When an object moves along a circular path with a **constant speed**, its motion is called **Uniform Circular Motion (UCM)**.

Simple Example: Think of a fan blade spinning, or a stone tied to a string whirled in a circle. The stone moves at the same speed, but its **direction keeps changing** at every point!

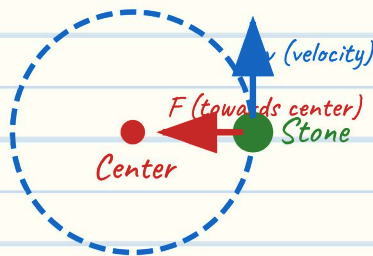
⚠️ **Tricky Point (Board loves this!):** In UCM, the **speed is constant** but the **velocity is NOT constant**. Why? Because **velocity = speed + direction**. Since direction keeps changing, velocity changes. And if velocity changes → there is **acceleration**!

3. Centripetal Force

Definition: The force that acts on an object moving in a circular path and is directed towards the center of the circle is called **Centripetal Force**. ("Centripetal" means "center-seeking")

Simple Example: Imagine you tie a stone to a rope and spin it around your head. Your hand keeps pulling the rope inward (towards the center). That inward pull = Centripetal Force!

If you let go of the string, the stone flies away in a straight line (tangent). This proves that centripetal force is **NEEDED** to keep circular motion going.



↑ Centripetal Force points **INWARD**, Velocity is always **TANGENT**

Formula for Centripetal Force:

$$F = mv^2 / r$$

- **F** = Centripetal Force (in Newtons, N)
- **m** = Mass of the object (in kg)
- **v** = Speed of the object (in m/s)
- **r** = Radius of the circular path (in m)



Real Life Examples of Centripetal Force:

-  Car turning on a curved road → friction provides centripetal force.
-  Earth orbiting the Sun → gravity provides centripetal force.
-  Moon orbiting the Earth → gravity provides centripetal force.

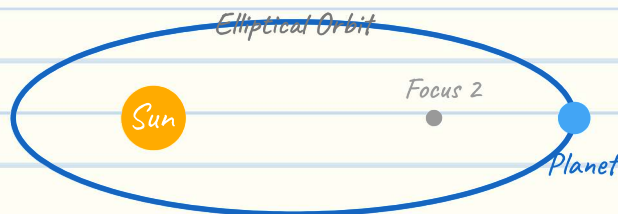
4. Kepler's Laws of Planetary Motion 🪐

A scientist named *Johannes Kepler* studied how planets move and gave 3 important laws.

Law 1: Law of Orbits 🌐

Statement: The orbit of a planet is an **ellipse** (oval shape) with the **Sun at one of its foci**.

Simple Words: Planets don't go in a perfect circle. They go in a slightly squashed circle called an ellipse. The Sun is not exactly in the middle – it's off to one side!



Law 2: Law of Areas 📐

Statement: The line joining a planet and the Sun sweeps **equal areas in equal intervals of time**.

Simple Words: Imagine a string from the Sun to a planet. As the planet moves, it "paints" an area. The planet paints equal sized areas in equal time.

This means: Planet is **closer to Sun** → **moves FASTER**. Farther from Sun → **moves SLOWER**.



Law 3: Law of Periods 🕒

Statement: The square of the time period (T) of revolution is directly proportional to the cube of the average distance (r) from the Sun.

$$T^2 \propto r^3 \text{ or } T^2/r^3 = \text{constant}$$

Simple Words: Planets far away from the Sun take much longer to go around it.

5. Newton's Universal Law of Gravitation

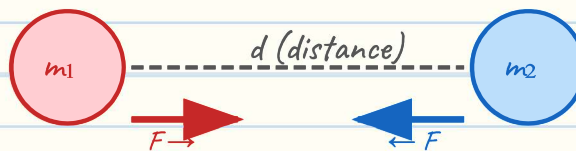
The Story: Sir Isaac Newton saw an apple fall and thought – "The same force that pulls the apple must be pulling the Moon towards Earth!" This led him to the Universal Law of Gravitation.

Statement: Every object in the universe attracts every other object with a force which is:

- (i) *directly proportional* to the product of their masses, and
- (ii) *inversely proportional* to the square of the distance between them.

The Formula:

$$F = G \times (m_1 \times m_2) / d^2$$



Both objects pull EACH OTHER with EQUAL force (opposite direction!)

What does each letter mean?

Symbol	Meaning	Unit
F	Gravitational Force	Newton (N)
G	Universal Gravitational Constant	Nm^2/kg^2
m_1, m_2	Masses of the two objects	Kilogram (kg)
d	Distance between their centers	Metre (m)

About 'G' (Gravitational Constant):

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

- G was first measured by scientist **Henry Cavendish**.
- G is a **universal constant** – same everywhere in the universe!
- Its value does **NOT** change with place, time, or temperature.

6. Acceleration due to Gravity (g)

Definition: The acceleration produced in a freely falling body due to the gravitational force of the Earth is called **acceleration due to gravity**. It is denoted by ' g '.

Simple Words: When you drop a ball, it falls faster and faster (it accelerates). This "speeding up" is caused by Earth's gravity. The rate at which it speeds up is called g .

On Earth's surface: $g \approx 9.8 \text{ m/s}^2$

This means: Every second, a falling object's speed increases by 9.8 m/s !

★ **Derivation: Relation between g and G**



Derivation: $g = GM/R^2$

Let's find the value of ' g ' using Newton's Law:

- 1 Consider an object of mass m on Earth's surface.
- 2 Earth's mass = M , Earth's radius = R
- 3 By Newton's Gravitation Law, force on the object:

$$F = GMm / R^2$$

- 4 By Newton's 2nd Law ($F = ma$), for this object:

$$F = mg \quad (\text{where } a = g)$$

- 5 Equating both:

$$mg = GMm / R^2$$

- 6 Cancel ' m ' from both sides:

$$g = GM / R^2$$



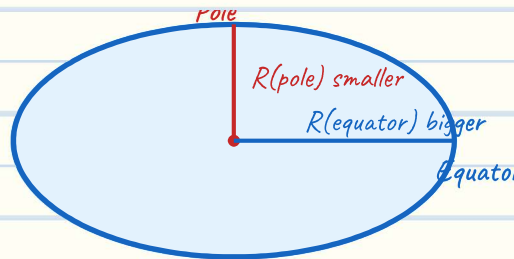
Notice: The value of ' g ' does NOT depend on the mass (m) of the falling object! Whether it's an ant or an elephant, ' g ' is the same! It depends only on Earth's mass (M) and radius (R).

7. Variation of 'g'

The value of 'g' is NOT the same everywhere. It changes based on 3 things:

A) Variation with Shape of Earth

Earth is NOT a perfect sphere! It is slightly **flat at the poles** and **bulging at the equator**.



- Since $g = GM/R^2$, and R is **smaller at poles** → g is **MAXIMUM** at the poles.
- R is **bigger at equator** → g is **MINIMUM** at the equator.

g at poles (9.832 m/s^2) > g at equator (9.78 m/s^2)

B) Variation with Height (Altitude)

As you go higher above Earth's surface (airplane, rocket), you move farther from Earth's center.

- Distance from center increases → g **DECREASES**.
- At very large height, g becomes almost zero!

Simple Memory: Go UP  = g goes DOWN 

C) Variation with Depth

If you dig a deep hole going towards the center of Earth:

- As you go deeper, Earth's mass below you decreases.
- So the pull on you decreases → g **DECREASES**.
- At the exact center of Earth: $g = 0$ (zero gravity!)

Simple Memory: Go DOWN  = g ALSO goes DOWN 

 **Board Exam Favourite:** "Where is 'g' maximum?" → Answer: **At the poles, on Earth's surface**. It decreases if you go up OR down from the surface!

8. Mass vs. Weight

Very Important! This "difference" question comes almost every year!

What is Mass?

Definition: Mass is the amount of matter contained in an object. It is a **scalar quantity**.

What is Weight?

Definition: Weight is the force with which the Earth attracts an object towards its center. It is a **vector quantity**.

The BIG Difference Table:

Feature	MASS	WEIGHT
What is it?	Amount of matter in you	How hard a planet PULLS you
Type	Scalar (magnitude only)	Vector (magnitude + direction)
Formula	m	$W = m \times g$
SI Unit	Kilogram (kg)	Newton (N)
Changes?	NO! Same everywhere	YES! Changes with 'g'
Can be zero?	Never zero	Yes, at center of Earth
Measured with	Physical balance	Spring balance

 **Example:** A person has mass = 60 kg.

- Weight on Earth = $60 \times 9.8 = 588 \text{ N}$
- Weight on Moon = $60 \times 1.63 = 97.8 \text{ N}$ (Moon's $g \approx 1/6$ th of Earth!)
- Mass on Moon = **Still 60 kg!** (Mass NEVER changes!)

 **Memory Trick:**

Mass = My Matter $\rightarrow M \ \& \ M \rightarrow$ never changes! 

Weight = World's pull $\rightarrow W \ \& \ W \rightarrow$ changes with the world/planet!

9. Free Fall

Definition: When an object falls **only** under the influence of gravitational force (with no air resistance), it is said to be in **free fall**.

Simple Words: Drop a ball in a room with **NO** air (vacuum). It falls only because of gravity. This is free fall!

! Key Point: In free fall, **ALL** objects (heavy or light) fall at the **SAME** rate! A feather and an iron ball reach the ground at the same time in a vacuum! (Proved by Galileo)

Equations of Motion for Free Fall:

In free fall: initial velocity $u = 0$ and acceleration $a = g$.

Normal Equation	Free Fall Version ($u=0, a=g$)
$v = u + at$	$v = gt$
$s = ut + \frac{1}{2}at^2$	$s = \frac{1}{2}gt^2$
$v^2 = u^2 + 2as$	$v^2 = 2gs$

! Sign Convention (Very Important!):

- Object falling **DOWN** $\rightarrow g = +9.8 \text{ m/s}^2$ (positive)
- Object thrown **UP** $\rightarrow g = -9.8 \text{ m/s}^2$ (negative)

Solved Numerical 1:

Q: A stone is dropped from height 20m. Find time to reach ground. ($g = 10 \text{ m/s}^2$)

Solution: Given: $s = 20\text{m}$, $u = 0$, $g = 10 \text{ m/s}^2$

Formula: $s = \frac{1}{2}gt^2 \rightarrow 20 = \frac{1}{2} \times 10 \times t^2 \rightarrow 20 = 5t^2 \rightarrow t^2 = 4$

$t = 2 \text{ seconds } \checkmark$

Solved Numerical 2:

Q: A ball is thrown upward with velocity 20 m/s. Find max height. ($g = 10 \text{ m/s}^2$)

Solution: Given: $u = 20 \text{ m/s}$, $v = 0$ (stops at top), $g = -10 \text{ m/s}^2$

Formula: $v^2 = u^2 + 2as \rightarrow 0 = 400 + 2(-10)(s) \rightarrow 0 = 400 - 20s$

$s = 20 \text{ metres } \checkmark$

 Solved Numerical 3:

Q: Calculate gravitational force between two objects of masses 50 kg and 100 kg placed 2m apart. ($G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$)

Solution:

Given: $m_1 = 50 \text{ kg}$, $m_2 = 100 \text{ kg}$, $d = 2 \text{ m}$

Formula: $F = G \times m_1 \times m_2 / d^2$

$$F = 6.67 \times 10^{-11} \times 50 \times 100 / (2)^2$$

$$F = 6.67 \times 10^{-11} \times 5000 / 4$$

$$F = 6.67 \times 10^{-11} \times 1250$$

$$F = 8.34 \times 10^{-8} \text{ N} \checkmark$$

 Solved Numerical 4:

Q: Calculate the value of 'g' on Earth.

($M = 6 \times 10^{24} \text{ kg}$, $R = 6.4 \times 10^6 \text{ m}$, $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$)

Solution:

Formula: $g = GM / R^2$

$$g = (6.67 \times 10^{-11} \times 6 \times 10^{24}) / (6.4 \times 10^6)^2$$

$$g = (40.02 \times 10^{13}) / (40.96 \times 10^{12})$$

$$g \approx 9.77 \text{ m/s}^2 \checkmark$$

 Solved Numerical 5:

Q: An object is dropped from a tower. It reaches the ground in 3 seconds. Find the height of the tower. ($g = 10 \text{ m/s}^2$)

Solution:

Given: $u = 0$, $t = 3 \text{ s}$, $g = 10 \text{ m/s}^2$

Formula: $s = \frac{1}{2} gt^2$

$$s = \frac{1}{2} \times 10 \times (3)^2 = 5 \times 9$$

$$s = 45 \text{ metres} \checkmark$$

 Solved Numerical 6:

Q: Weight of a 50 kg person on Earth and Moon?

($g_{\text{Earth}} = 9.8 \text{ m/s}^2$, $g_{\text{Moon}} = 1.63 \text{ m/s}^2$)

Solution: $W = mg$

$$\text{Earth: } 50 \times 9.8 = 490 \text{ N} \mid \text{Moon: } 50 \times 1.63 = 81.5 \text{ N} \mid \text{Mass} = 50 \text{ kg (same!)}$$

10. Gravitational Potential Energy

Definition: The energy stored in an object due to its position (height) in a gravitational field is called **Gravitational Potential Energy**.

Simple Words: When you lift a book and put it on a shelf, you do work against gravity. That work gets stored as energy. The higher you put it, the more energy!

$$\text{Potential Energy (U)} = m \times g \times h$$

- m = mass of the object (kg)
- g = acceleration due to gravity (m/s^2)
- h = height above the ground (m)



Example: A ball of mass 2 kg is on a table 3m high. PE = ?

$$U = mgh = 2 \times 9.8 \times 3 = 58.8 \text{ Joules}$$

11. Escape Velocity

Definition: The minimum velocity needed for an object to escape Earth's gravitational pull and never come back is called **Escape Velocity**.

Simple Words: If you throw a ball up, it always comes back. What if you throw it SO FAST that it leaves Earth forever? That speed is the Escape Velocity!

Formula:

$$v(\text{esc}) = \sqrt{2GM/R} = \sqrt{2gR}$$

Value for Earth:

$$\text{Escape Velocity of Earth} = 11.2 \text{ km/s} = 11,200 \text{ m/s}$$

That's about **40,320 km/hour!** That's how fast a rocket must go!



Fun Fact: The Moon has lower escape velocity (2.4 km/s) because it has less mass. That's why the Moon has almost no atmosphere – gas molecules easily escape!

12. Important Constants & Values

Keep these values ready for numericals!

Constant / Value	Symbol	Value
Gravitational Constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Gravity on Earth	g	9.8 m/s^2
Mass of Earth	M	$6 \times 10^{24} \text{ kg}$
Radius of Earth	R	$6.4 \times 10^6 \text{ m (6400 km)}$
Mass of Moon	—	$7.34 \times 10^{22} \text{ kg}$
Radius of Moon	—	$1.74 \times 10^6 \text{ m}$
Escape Velocity (Earth)	$v(\text{esc})$	11.2 km/s
g on Moon	—	$1.63 \text{ m/s}^2 (\approx g/6)$

13. All Formulas at a Glance

Concept	Formula
Centripetal Force	$F = mv^2/r$
Kepler's 3rd Law	$T^2 \propto r^3$
Newton's Gravitational Law	$F = Gm_1m_2/d^2$
Acceleration due to gravity	$g = GM/R^2$
Weight	$W = mg$
Free Fall (v)	$v = gt$
Free Fall (s)	$s = \frac{1}{2}gt^2$
Free Fall (v^2)	$v^2 = 2gs$
Gravitational PE	$U = mgh$
Escape Velocity	$v = \sqrt{2GM/R} = \sqrt{2gR}$



Tip: Photocopy this page and stick it on your study table. Revise these formulas every day before the exam!

Board Exam Cheat Sheet

Tips, Tricks & Memory Hacks

Memory Tricks:

- **O.A.T.** for Kepler → Orbit (Ellipse), Area (Equal), Time ($T^2 \propto r^3$)
- **M&M** for Mass → "My Matter" – Never changes! 
- **W&W** for Weight → "World's pull" – Changes with planet!
- "Go UP or DOWN, g goes DOWN" → max only on surface.
- "11.2" → Escape velocity. "11 to fly!" 
- "Centri-PETAL" → Petals point inward → Force to CENTER 

Commonly Asked Board Questions:

1. State Newton's Universal Law of Gravitation. (2 marks)
2. Distinguish between Mass and Weight. (2 marks) → **Draw table!**
3. Derive the relation $g = GM/R^2$. (3 marks)
4. State Kepler's three laws. (3 marks)
5. Explain variation of 'g' with altitude and depth. (3 marks)
6. What is escape velocity? State its formula. (2 marks)
7. Numerical: Gravitational force between objects. (3 marks)
8. Numerical: Free fall problems. (3 marks)
9. Why is 'g' maximum at poles? (2 marks)
10. Define centripetal force. Give examples. (2 marks)

Exam Day Tips:

-  **ALWAYS write units!** (N, kg, m/s^2 , m). Half marks cut if missed!
-  "Difference" questions → Draw a **TABLE**. Examiners love it!
-  Kepler's laws → Draw the **ellipse diagram**.
-  Numericals → Write **Given** → **Formula** → **Solution** → **Answer**.
-   Sign of g: Falling = +g, Thrown up = -g.
-  Derivation $g = GM/R^2$ is a **3-mark favourite**. Practice it!

 You've got this! All the best for your board exam! 

QUICK RECALL ONE-PAGER

Mnemonics • Tricks • Shortcuts – Stick This On Your Wall!

abc Trick 1: "OAT" – Kepler's 3 Laws

- **O** = Orbit → Planets move in Oval (Ellipse)
- **A** = Area → Equal Areas in equal time
- **T** = Time → $T^2 \propto r^3$

🍬 Trick 2: "M&M vs W&W" – Mass vs Weight

- Mass = My Matter = M&M → SAME everywhere! 🍬
- Weight = Which World? → CHANGES with planet! 🌍 🌙
- Mass → kg, Weight → Newton. "Weighing gives Newtons"

⬆️ ⬆️ Trick 3: "Everywhere Down" – Variation of g

- Go UP? → g goes DOWN ⬆️ / Go DOWN? → g goes DOWN ⬆️
- g is MAX on surface at POLES! Think: "Surface pe Pole = Superman!" 💪

🌸 Trick 4: "PETAL" – Centripetal Force

- Centri-PETAL → Petals point INWARD → Force towards CENTER 🌸
- $F = mv^2/r$ → "My Very Very Round path"

1234 Trick 5: Remembering Constants

- $G = 6.67 \times 10^{-11}$ → "6 baje 67 minute" power -11
- $g = 9.8$ → "Almost 10" (use 10 for quick calc!)
- Escape = 11.2 km/s → "112 = Police! Emergency escape!" 🚓 🚀
- Earth = 6×10^{24} kg / Radius = 6400 km

🎲 Trick 6: Derivation Shortcut – $g = GM/R^2$

- " $mg = GMm/R^2$ " → cancel 'm' → $g = GM/R^2$ – Only 2 steps!
- "Newton's 2nd Law MEETS Gravity Law" = g formula!

📄 Trick 7: Free Fall Sign Rule

- Falling DOWN → $g = +ve$ ✅ / Thrown UP → $g = -ve$ ❌
- "Gravity HAPPY falling (+), SAD going up (-)" 😊 😞

ALL FORMULAS – BIG & CLEAR

Print this page. Stick it on your wall. Revise daily!

Newton's Universal Law of Gravitation

$$F = G \times m_1 \times m_2 / d^2$$

Acceleration due to Gravity

$$g = GM / R^2$$

Weight of an Object

$$W = m \times g$$

Centripetal Force

$$F = mv^2 / r$$

Escape Velocity

$$v = \sqrt{2GM/R} = \sqrt{2gR}$$

= 11.2 km/s for Earth 🚀

Kepler's Third Law

$$T^2 \propto r^3$$

Free Fall Equations ($u = 0$, $a = g$)

$$v = gt \quad | \quad s = \frac{1}{2}gt^2 \quad | \quad v^2 = 2gs$$

Gravitational Potential Energy

$$U = m \times g \times h$$